

Project Design Phase
Solution Architecture

Date	9 th march 2026
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture defines how the proposed system solves the identified problem using appropriate technologies and system components. It provides a clear view of the structure, components, data flow, and interactions between different modules in the system.

The proposed solution architecture for the Real-Time Healthcare Anomaly Detection System integrates streaming technologies, machine learning models, backend services, and visualization tools to monitor patient health data continuously and detect anomalies in real time.

The system architecture is designed to be scalable, reliable, and capable of processing continuous healthcare data streams.

Solution Architecture Diagram:

AI-Driven Healthcare Anomaly Detection System Architecture

1 Patient Sensors / Data Generator

Heart Rate, SpO₂, Temperature, Blood Pressure

HTML5, CSS3, JS, Jinja2

Real-Time Data Streams

2 Apache Kafka (by Confluent)

kafka, TensorFlow, Isolation Forest

Anomaly Scores & Alerts

3 ML Inference Engine (Autoencoder + Isolation Forest)

Reprocessed Data → Recosultiers → Anomaly Score (LOW, MEDIUM, HIGH)

Logs Historical Data

4 Flask Backend

PostgreSQL, Flask, Web Dashboard

Anomaly Scores & Alerts

5 Data Storage & Backend Logic

PostgreSQL, Email Alerts (SMTP)

- 5 Layer 1: Data Ingestion** (Sensors)
- 4 Layer 2: Stream Processing**
- 3 Layer 3: Machine Learning Inference Engine** (Autoencoder + Isolation Forest)

Reference:

[1] A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, O'Reilly Media, 2019.

[2] F. T. Liu, K. M. Ting, and Z. Zhou, "Isolation Forest," IEEE ICDM, 2008.

[3] Apache Kafka Documentation. <https://kafka.apache.org>